

LAN Security: Firewalls, Access Control, and Hardening

Hosts that share a local area network can often reach one another without crossing an Internet firewall. Lateral movement is therefore a first-class risk. Defenses start with physical control of switch closets and continue with port security, 802.1X authentication, and network access control that checks a laptop before it receives a productive VLAN.

Access control lists on layer-3 boundaries restrict which subnets may connect to management interfaces, file servers, and building automation. A host-based firewall on each workstation adds defense in depth. Unused services should be disabled so that a compromised printer cannot scan the entire campus.

Switch hardening includes a secure management VLAN, encrypted administration protocols, disabled unused ports, and storm control against broadcast floods. DHCP snooping and dynamic ARP inspection reduce spoofing on access switches. Logging should be sent to a collector so that port flaps and authentication failures are visible.

Encryption protects data that leaves the building, but inside the LAN many protocols remain cleartext unless administrators insist on TLS and SSH. Regular reviews of firewall rules, VLAN maps, and user accounts keep the local network from slowly returning to a single trusted soup of devices.